

Question:
A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?
Answer Choices:
A. Epstein-Barr virus (EBV)
B. Measles virus
C. Human immunodeficiency virus (HIV)
D. Human papillomavirus

Reasoning + Answer:

- <think>
- CD 2 1 is a receptor protein on the surface of B cells , primarily involved in the interaction between B cells and T helper cells .
 - B cells that do not express CD 2 1 would have a reduced ability to interact with T helper cells .
 - EBV and measles virus both rely on B cells for their pathogenesis as they can infect B cells and use them as a reservoir or for replication .
 - HIV primarily targets CD 4 + T cells , not B cells , and does not rely on B cells for its pathogenesis .
 - Human papillomavirus (HPV) infects epithelial cells and does not involve B cells in its pathogenesis .

Given these points , the genetic modification of B cells to not express CD 2 1 would most likely affect the pathogenesis of EBV and measles virus , as these rely on B cells for their life cycle . However , since EBV specifically uses B cells for its life cycle and replication , it would be more directly affected than measles virus , which can infect B cells but does not rely on them as extensively for its pathogenesis .

- Answer Choices :
- A . Epstein -B arr virus (EB V)
 - B . Me as les virus
 - C . Human immunodeficiency virus (HIV)
 - D . Human pap ill om avirus

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< answer >A . Epstein -B arr virus (EB V)</ answer >